



B.S. 4 Years Program / Fifth Semester – Fall 2024

Paper: Sampling Techniques (Theory)

Course Code: STAT-305

Roll No.

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- i) Define probability and non-probability sampling? Which of them is more preferable?
- ii) Define mean square error and its uses in sampling theory.
- iii) What is stratified sampling? What are its advantages?
- iv) Write a note on inverse sampling.
- v) What are the points to be considered while estimating the sample size?
- vi) Differentiate between linear and circular systematic sampling.

Answer the following questions.

(3x10=30)

Q.2. For simple random sampling, show that the sample mean, $\bar{y}$ is an unbiased estimator of the population mean $\bar{Y}$ and variance of sample mean is given by

$$V(\bar{y}) = \left(\frac{N-1}{Nn} \right) S^2$$

Use Cornfield approach to prove this.

Q.3. In stratified random sampling, show that the estimator $\bar{y}_s = \sum_{h=1}^L W_h \bar{y}_h$ is an unbiased estimator of the

population mean $\bar{Y}$ with variance given by: $Var(\bar{y}_s) = \sum_{h=1}^L \frac{W_h^2 S_h^2}{n_h} (1 - f_h)$

Q.4. Show that the mean of a systematic sample is more precise than the mean of a simple random sample if and only if $S_{wy}^2 > S^2$